Physics informed neural network of a moving conductive string in a magnetic field

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1 Physical Model

Consider an infinite conductive string an a homogeneous magnetic field, $\mathbf{B}_0 = B_0\mathbf{i}_x$, where B_0 is a constant. The string is being pulled at a constant velocity, $\mathbf{V} = V_0\mathbf{i}_x$, through two rings which are located a distance l from each other. The motion of the string is considered in a fixed Cartesian coordinate system (x, y, z), with unit vectors $\mathbf{i}_x, \mathbf{i}_y, \mathbf{i}_z$. The center of one of the rings is assumed to be the center of the coordinate system.

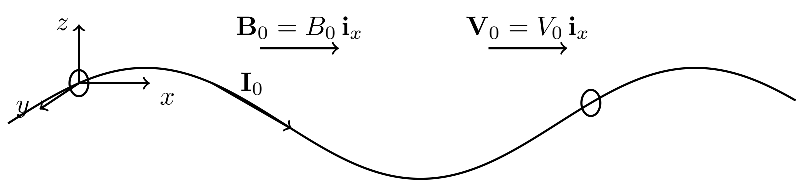

Figure 1: Schematic of the string with the applied fields and forces.

The influence of gravity and other external forces are neglected. The unit length and the position of each material point of the string is assumed to be constant. We identify the position of each material point of the string through its arc length $s \in [0, l]$. In Euclidean space, we denote the displacement of the string by $\mathbf{r}(s,t) = x(s,t)\mathbf{i}_x + y(s,t)\mathbf{i}_y + z(s,t)\mathbf{i}_z$. We also assume that $\mathbf{k} = \frac{\partial x}{\partial s}\mathbf{i}_x + \frac{\partial y}{\partial s}\mathbf{i}_y + \frac{\partial z}{\partial s}\mathbf{i}_z$ represents the unit vector directed along the tangent to the deformed line of the string. If we assume a current $\mathbf{I} = I_0(t)\mathbf{k}$ flows through the string, then the magnetic field will exert forces on the string. In this particular case, the forces acting on the string are called Laplace forces given by

$$\mathbf{F} = I_0 B_0(\mathbf{k} \times \mathbf{i}_x). \tag{1}$$

The motion of the string is described by

$$\frac{\partial \mathbf{Q}}{\partial s} + \mathbf{F} = \rho\mathbf{R}, \qquad \mathbf{Q} \times \mathbf{k} = 0, \qquad (\frac{\partial \mathbf{r}}{\partial s}) = 1, \tag{2}$$

where $\mathbf{Q}$ is the internal tension in the string, ρ is the mass density per unit length, and $\mathbf{R}$ is the inertial force of the string given by

$$R_x = 0, \qquad R_y = \frac{\partial^2 y}{\partial t^2} + 2V_0\frac{\partial^2 y}{\partial s \partial t} + V_0^2\frac{\partial^2 y}{\partial s^2} + \eta\frac{\partial y}{\partial t}, \qquad R_z = \frac{\partial^2 z}{\partial t^2} + 2V_0\frac{\partial^2 z}{\partial s \partial t} + V_0^2\frac{\partial^2 z}{\partial s^2} + \eta\frac{\partial z}{\partial t}, \tag{3}$$

where η is a damping coefficient. The inertial force in the x-direction is zero since we assume that the string is being pulled in the x-direction with a constant velocity. The complex function

$u(s,t) = y(s,t) + iz(s,t)$ will be used to represent the displacement of the string in the y and z directions.

Using the aforementioned equations, we can obtain the governing equations for lateral vibrations

$$\rho\left[u_{tt} + \eta u_t + 2V_0 u_{st} + V_0^2 u_{ss}\right] = \frac{\partial}{\partial s}\left(\frac{P_0\, u_s}{\sqrt{1-|u_s|^2}}\right) - iI_0 B_0 u_s, \tag{4}$$

where the subscripts represent partial derivatives, $0 < s < l$, and $P_0 = Q_x$ is the tension applied to the string at rest. The boundary conditions are given by

$$u(0,t) = 0, \qquad u(l,t) = 0, \qquad t \geq 0. \tag{5}$$

These boundary conditions specify that the string is fixed at the rings. Initial conditions will be assigned appropriately. if we assume that the deformations are negligibly small, i.e., $|u_s| \ll 1$, then the governing equation can be linearized as

$$\frac{\rho}{P_0}\left[u_{tt} + \eta u_t + 2V_0 u_{st}\right] = \left(1 - \frac{\rho V_0^2}{P_0}\right)u_{ss} - i\frac{I_0 B_0}{P_0}u_s, \tag{6}$$

where $0 < s < l$ and $t \geq 0$.

2 Physics-Informed Neural Network Formulation

To approximate solutions of the governing equation (**??**), we construct a physics-informed neural network (PINN) that represents the transverse displacement of the string.

2.1 Neural Network Representation

Let

$$\hat{u}(s,t;\boldsymbol{\theta}) = \hat{y}(s,t;\boldsymbol{\theta}) + i\,\hat{z}(s,t;\boldsymbol{\theta})$$

denote the neural network approximation of the complex displacement

$$u(s,t) = y(s,t) + iz(s,t),$$

where $\boldsymbol{\theta}$ represents the trainable parameters (weights and biases) of the network. In practice, the network outputs the two real-valued components $(\hat{y}, \hat{z})$ simultaneously.

The network takes (s,t) as inputs and consists of fully connected layers with smooth activation functions (e.g., tanh) so that spatial and temporal derivatives can be computed via automatic differentiation.

2.2 Dimensionless Form

For numerical stability and to reduce the number of physical parameters, we use the nondimensional variables

$$x = \frac{s}{l}, \qquad \tau = \frac{c_0 t}{l}, \qquad c_0 = \sqrt{\frac{P_0}{\rho}},$$

and define

$$\beta = \frac{V_0}{c_0}, \qquad \mu = \frac{\eta l}{c_0}, \qquad \Gamma = \frac{I_0 B_0 l}{P_0}.$$

In these variables, the linearized governing equation (**??**) becomes

$$u_{\tau\tau} + \mu u_\tau + 2\beta u_{x\tau} + \beta^2 u_{xx} = u_{xx} - i\Gamma u_x. \tag{9}$$

Separating real and imaginary parts yields the coupled system

$$y_{\tau\tau} + \mu y_\tau + 2\beta y_{x\tau} + (\beta^2 - 1)y_{xx} - \Gamma z_x = 0, \tag{7}$$

$$z_{\tau\tau} + \mu z_\tau + 2\beta z_{x\tau} + (\beta^2 - 1)z_{xx} + \Gamma y_x = 0. \tag{8}$$

2.3 PDE Residual

The PINN enforces the governing equations by minimizing the residuals

$$R_y(x,\tau;\boldsymbol{\theta}) = \hat{y}_{\tau\tau} + \mu\hat{y}_\tau + 2\beta\hat{y}_{x\tau} + (\beta^2 - 1)\hat{y}_{xx} - \Gamma\hat{z}_x, \tag{9}$$

$$R_z(x,\tau;\boldsymbol{\theta}) = \hat{z}_{\tau\tau} + \mu\hat{z}_\tau + 2\beta\hat{z}_{x\tau} + (\beta^2 - 1)\hat{z}_{xx} + \Gamma\hat{y}_x. \tag{10}$$

These derivatives are computed using automatic differentiation.

2.4 Boundary and Initial Conditions

The boundary conditions (**??**) become

$$\hat{y}(0,\tau) = \hat{y}(1,\tau) = 0, \qquad \hat{z}(0,\tau) = \hat{z}(1,\tau) = 0.$$

Initial conditions are imposed as

$$\hat{y}(x,0) = y_0(x), \quad \hat{z}(x,0) = z_0(x),$$

$$\hat{y}_\tau(x,0) = v_{y,0}(x), \quad \hat{z}_\tau(x,0) = v_{z,0}(x).$$

2.5 Loss Function

Let $\{(x_r^i, \tau_r^i)\}_{i=1}^{N_r}$ denote interior collocation points, $\{\tau_b^i\}_{i=1}^{N_b}$ boundary points, and $\{x_0^i\}_{i=1}^{N_0}$ initial points. The total loss is defined as

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N_r}\sum_{i=1}^{N_r}\left(|R_y(x_r^i,\tau_r^i;\boldsymbol{\theta})|^2 + |R_z(x_r^i,\tau_r^i;\boldsymbol{\theta})|^2\right) \tag{11}$$

$$+ \frac{1}{N_b}\sum_{i=1}^{N_b}\left(|\hat{y}(0,\tau_b^i)|^2 + |\hat{y}(1,\tau_b^i)|^2 + |\hat{z}(0,\tau_b^i)|^2 + |\hat{z}(1,\tau_b^i)|^2\right) \tag{12}$$

$$+ \frac{1}{N_0}\sum_{i=1}^{N_0}\left(|\hat{y}(x_0^i,0) - y_0(x_0^i)|^2 + |\hat{z}(x_0^i,0) - z_0(x_0^i)|^2\right) \tag{13}$$

$$+ \frac{1}{N_0}\sum_{i=1}^{N_0}\left(|\hat{y}_\tau(x_0^i,0) - v_{y,0}(x_0^i)|^2 + |\hat{z}_\tau(x_0^i,0) - v_{z,0}(x_0^i)|^2\right). \tag{14}$$

The neural network parameters $\boldsymbol{\theta}$ are optimized using gradient-based methods (e.g., Adam) to minimize $\mathcal{L}$.

2.6 Static PINN for Buckling Equilibria

To study static buckling equilibria, we consider time-independent solutions of (**??**). Setting all time derivatives to zero yields

$$(\beta^2 - 1)u_{xx} + i\Gamma u_x = 0.$$

Splitting into real and imaginary parts gives

$$(\beta^2 - 1)y_{xx} - \Gamma z_x = 0, \tag{15}$$

$$(\beta^2 - 1)z_{xx} + \Gamma y_x = 0, \tag{16}$$

subject to

$$y(0) = y(1) = 0, \qquad z(0) = z(1) = 0.$$

A second PINN is constructed with input x only and outputs $(\hat{y}(x), \hat{z}(x))$. The residuals of the above system are minimized together with the boundary conditions. Because the trivial solution $(y, z) = (0, 0)$ always satisfies the equations, a small symmetry-breaking constraint (e.g., $\hat{y}(1/2) = A$) is added to select a nontrivial buckled branch.